

Chap 2. The Space of Rough Paths

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2. Main contents
 - Basic definitions
 - The space of α -Hölder rough paths
 - The space of geometric rough paths
 - Rough paths as Lie group valued paths
3. Conclusion

Review

Review: Definitions

Let $\alpha > 0$ and E be a Banach space.

Definition. A function $X : [0, T] \rightarrow E$ is α -Hölder continuous if

$$\|X\|_\alpha := \sup_{s \neq t \in [0, T]} \frac{|X_{s,t}|_E}{|t - s|^\alpha} < \infty.$$

The space of all α -Hölder continuous functions from $[0, T]$ into E is denoted by $\mathcal{C}^\alpha([0, T]; E) = \mathcal{C}^\alpha(E) = \mathcal{C}^\alpha$.

Notation.

1. $X_{s,t} := X_t - X_s$ is the path increment.
2. For a function $\Xi : [0, T]^2 \rightarrow E$, we write $\Xi \in \mathcal{C}_2^\alpha([0, T]^2; E)$ if

$$\|\Xi\|_\alpha := \sup_{s \neq t \in [0, T]} \frac{|\Xi_{s,t}|_E}{|t - s|^\alpha} < \infty, \quad \text{where} \quad \Xi_{s,t} := \Xi(s, t).$$

Remark.

1. If $X \in \mathcal{C}^\alpha([0, T]; E)$, then its increment $X_{s,t} \in \mathcal{C}_2^\alpha([0, T]^2; E)$.
2. The quantity $\|\cdot\|_\alpha$ is a semi-norm; however, the quantity

$$\|X\|_{\mathcal{C}_\alpha} := |X_0|_E + \|X\|_\alpha,$$

is a norm that makes $\mathcal{C}^\alpha(E)$ a Banach space (in general not separable).

3. If $\alpha < \beta$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

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- * When working with paths starting at the origin, the term $|X_0|_E$ can be omitted: we can work directly with $\|\cdot\|_\alpha$.
 - * The same is true if we are interested in the α -Hölder metric between two paths started at the same point.
3. If $\alpha < \beta$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

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- Main object: differential equation of the form

$$dY_t = f(Y_t)dX_t. \quad (1)$$

- E and V : Banach spaces
- $X : [0, T] \rightarrow E$, input; $Y : [0, T] \rightarrow V$, output (unknown)
- $f : V \rightarrow \mathcal{L}(E; V)$ is a smooth function
- $\mathcal{L}(E; V) :=$ the space of continuous linear maps from E to V .

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 - $\mathcal{L}(E; V) :=$ the space of continuous linear maps from E to V .
- * Common choice : $E = \mathbb{R}^d$ and $V = \mathbb{R}^n$, so that $f : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$.
- * We expect X and Y to be continuous functions.

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$$Y_t = y_0 + \int_0^t f(Y_s)dX_s. \quad (2)$$

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- $X, Y \in \mathcal{C}^\alpha$ (and hence $t \mapsto f(Y_t)$ is α -Hölder as well)

Remark.

1. If $\alpha + \alpha > 1$, we can interpret the integral appearing in (2) as a Young integral and find the solution in the space $\mathcal{C}^\alpha$.
2. If $\alpha + \alpha \leq 1$, the question is more tricky; the theory of rough paths provides a convenient answer and many classical results about stochastic differential equations are recovered in rough paths theory.

- Suppose that $X \in \mathcal{C}^\alpha([0, T]; \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$,
and that $Y : [0, T] \rightarrow \mathbb{R}^n$ solves (1) with smooth $f : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$:

$$dY_t = f(Y_t)dX_t.$$

- We expect that, at least on small scales,

$$Y_{s,t} = f(Y_s)X_{s,t} + R(s, t),$$

where R is some remainder; and thus

$$f(Y_t) - f(Y_s) = Df(Y_s)Y_{s,t} + \tilde{R}(s, t) = Df(Y_s)f(Y_s)X_{s,t} + \hat{R}(s, t)$$

for suitable remainder terms $\tilde{R}$ and $\hat{R}$.

- Neglecting the remainder and setting $Z_s := Df(Y_s)f(Y_s)$, we have

$$\begin{aligned}\int_0^t f(Y_s) dX_s &= \int_0^t f(Y_0) + (f(Y_s) - f(Y_0)) dX_s \\ &\approx f(Y_0)X_{0,t} + Z_0 \int_0^t X_{0,s} \otimes dX_s.\end{aligned}$$

Note. Since $\alpha \leq \frac{1}{2}$, this is again an ill-posed task: Rough path theory proposes to *postulate* the value of

$$\int_s^t X_{s,r} \otimes dX_r := \mathbb{X}_{s,t}, \quad (*)$$

and to regard a path as not only X but a couple $(X, \mathbb{X})$.

Basic Definitions

The space of α -Hölder rough paths

Definition. For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ and a Banach space V , the space $\mathcal{C}^\alpha([0, T], V)$ of α -Hölder rough paths is the space of pairs

$$\mathbf{X} = (X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T]; V) \oplus \mathcal{C}_2^{2\alpha}([0, T]^2; V \otimes V),$$

satisfying the algebraic relation (Chen's relation):

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0, T].$$

Algebraic condition

- Chen's relation:

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0, T].$$

Remark.

1. Since $X_{t,t} = X_t - X_t = 0$, $\mathbb{X}_{t,t} = 0$, for all t .
2. This algebraic condition is natural:

If X is a smooth function and we read $(*)$ from right to left, that is,

$$\mathbb{X}_{s,t} := \int_s^t X_{s,r} \otimes dX_r \left(= \int_s^t X_{s,r} \otimes \dot{X}_r dr \right),$$

then it follows that

$$\begin{aligned} & \mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} \\ &= \int_s^t X_{s,r} \otimes dX_r - \int_s^u X_{s,r} \otimes dX_r - \int_u^t X_{u,r} \otimes dX_r \\ &= X_{s,u} \otimes \int_u^t dX_r = X_{s,u} \otimes X_{u,t}. \end{aligned}$$

Q. If X is assumed to be α -Hölder continuous, then what is the natural analytic condition that should be imposed on $\mathbb{X}$?

A. Since X is α -Hölder continuous,

$$|X_{s,t}| \lesssim |t-s|^\alpha \quad \Rightarrow \quad X_{\lambda s, \lambda t} \sim \lambda^\alpha X_{s,t}.$$

Considering (*), it is natural that $\mathbb{X}$ is also self-similar with

$$\mathbb{X}_{\lambda s, \lambda t} \sim \lambda^{2\alpha} \mathbb{X}_{s,t}.$$

• Analytic condition:

$$\|X\|_\alpha := \sup_{x \neq s \in [0, T]} \frac{|X_{s,t}|}{|t-s|^\alpha} < \infty, \quad \|\mathbb{X}\|_{2\alpha} := \sup_{x \neq s \in [0, T]} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} < \infty.$$

- If we ignore Chen's relation, the space $\mathcal{C}^\alpha$ is naturally embedded in the Banach space $\mathcal{C}^\alpha \oplus \mathcal{C}_2^{2\alpha}$ with (semi)-norm $\|X\|_\alpha + \|\mathbb{X}\|_{2\alpha}$.

However, due to the nonlinear constraint given by Chen's relation, the space $\mathcal{C}^\alpha$ is a closed set in $\mathcal{C}^\alpha \oplus \mathcal{C}_2^{2\alpha}$ but is not a linear space.

- On $\mathcal{C}^\alpha([0, T], V)$, we define the rough path norm

$$|||\mathbf{X}|||_\alpha := \|X\|_\alpha + \sqrt{\|\mathbb{X}\|_{2\alpha}},$$

which is not a norm in the common sense (the space is not linear), but is homogeneous with respect to the natural scaling $(X, \mathbb{X}) \mapsto (\lambda X, \lambda^2 \mathbb{X})$.

- Given rough paths $\mathbf{X}, \mathbf{Y} \in \mathcal{C}^\alpha([0, T], V)$, we define the (inhomogeneous) α -Hölder rough path metric

$$\varrho_\alpha(\mathbf{X}, \mathbf{Y}) := \sup_{s \neq t \in [0, T]} \frac{|X_{s,t} - Y_{s,t}|_V}{|t - s|^\alpha} + \sup_{s \neq t \in [0, T]} \frac{|\mathbb{X}_{s,t} - \mathbb{Y}_{s,t}|_{V \otimes V}}{|t - s|^{2\alpha}}.$$

Remark.

1. $(\mathcal{C}^\alpha, \varrho_\alpha)$ is a complete metric space.
2. If $\frac{1}{3} < \alpha < \beta \leq \frac{1}{2}$, then $\mathcal{C}^\beta \subset \mathcal{C}^\alpha$.

1. Given an arbitrary path $X \in \mathcal{C}^\alpha(V)$, it doesn't seem obvious that this path can indeed be lifted to a rough path $(X, \mathbb{X})$.

Lyons-Victoir Extension Theorem:

- If $\frac{1}{3} < \alpha < \frac{1}{2}$, given $X \in \mathcal{C}^\alpha(V)$, there is an associated rough path $(X, \mathbb{X})$. But the choice of $\mathbb{X}$ is not unique.
- For $\alpha = \frac{1}{2}$, it holds only when V is finite-dimensional.

2. Non-uniqueness of $\mathbb{X}$:

Chen's relation is not enough to determine the unique $\mathbb{X}$ for X :

$\therefore$ If we set $\tilde{\mathbb{X}}_{s,t} := \mathbb{X}_{s,t} + F_t - F_s$ for any continuous F , then pair $(X, \tilde{\mathbb{X}})$ satisfies Chen's relation.

$\mathbb{X}$ is determined up to increments of a 2α -Hölder continuous function F .

In general, there is no “canonical” choice for F .

3. Knowledge of the path $t \mapsto (X_{0,t}, \mathbb{X}_{0,t})$ determines the entire second order process $\mathbb{X}_{s,t}$.

If $(X, \mathbb{X})$ and $(X, \tilde{\mathbb{X}})$ are both rough paths, then by Chen's relation we have

$$\mathbb{X}_{s,t} - \tilde{\mathbb{X}}_{s,t} = \mathbb{X}_{0,t} - \tilde{\mathbb{X}}_{0,t} - \mathbb{X}_{0,s} + \tilde{\mathbb{X}}_{0,s}.$$

The space of geometric rough paths

- Suppose that X is smooth function taking values in $\mathbb{R}^d$ and set $\mathbb{X}_{s,t} := \int_s^t X_{s,r} \otimes dX_r$. Then, we have

$$\begin{aligned}\mathbb{X}_{s,t}^{i,j} + \mathbb{X}_{s,t}^{j,i} &= \int_s^t X_{s,r}^i dX_r^j + \int_s^t X_{s,r}^j dX_r^i \\ &= \int_s^t d(X^i X^j)_r - X_s^i X_{s,t}^j - X_s^j X_{s,t}^i \\ &= (X^i X^j)_{s,t} - X_s^i X_{s,t}^j - X_s^j X_{s,t}^i = X_{s,t}^i X_{s,t}^j\end{aligned}$$

and the symmetric part of $\mathbb{X}$ is given by

$$\text{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t}. \quad (4)$$

Definition. The set of geometric rough paths $\mathcal{C}_g^\alpha$ consists of those $(X, \mathbb{X}) \in \mathcal{C}^\alpha$ that satisfy equation (4).

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Remark

1. Equation (4) holds for smooth paths considered as rough paths with their canonical lift $\mathbb{X}_{s,t} = \int_s^t X_{s,r} \otimes dX_r$.

$\mathcal{C}_g^{0,\alpha}$: the closure of the set of these canonical lifts of smooth paths in $\mathcal{C}^\alpha$

- 2 . $\mathcal{C}_g^{0,\alpha}$ is strictly smaller than $\mathcal{C}_g^\alpha$.
- If $\frac{1}{3} < \alpha < \beta \leq \frac{1}{2}$, then $\mathcal{C}_g^\beta \subset \mathcal{C}_g^{0,\alpha}$.

Rough paths as Lie-group valued paths

- It is natural to see an α -rough path $(X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T], \mathbb{R}^d)$, $\alpha \in (\frac{1}{3}, \frac{1}{2}]$ as a function

$$\begin{aligned} [0, T]^2 &\longmapsto \mathbb{T}^{(2)} = \mathbb{T}^{(2)}(\mathbb{R}^d) := \mathbb{R} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \\ (s, t) &\longrightarrow (1, X_{s,t}, \mathbb{X}_{s,t}) =: \mathcal{X}_{s,t}. \end{aligned}$$

- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a, b, c) \otimes (a', b', c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

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Check. $(1, 0, 0)$: unit element and

$$(1, X_{s,t}, \mathbb{X}_{s,t}) = (1, X_{s,u}, \mathbb{X}_{s,u}) \otimes (1, X_{u,t}, \mathbb{X}_{u,t}), \quad \forall s, u, t \in [0, T].$$

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($\because X_{s,t} = X_{s,u} + X_{u,t}$ and Chen's relation)

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- Define a product $\otimes$ on $\mathbb{T}^{(2)}$ as

$$(a, b, c) \otimes (a', b', c') = (aa', ab' + a'b, ac' + a'c + b \otimes b').$$

- Chen's relation:

$$\mathcal{X}_{s,t} = \mathcal{X}_{s,u} \otimes \mathcal{X}_{u,t}.$$

Rough paths as Lie-group valued paths

► Set

$$\mathbb{T}_a^{(2)}(\mathbb{R}^d) := \{a\} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d)$$

Check. For each $(1, b, c) \in \mathbb{T}_1^{(2)}(\mathbb{R}^d)$,

$$(1, b, c) \otimes (1, -b, -c + b \otimes b) = (1, -b, -c + b \otimes b) \otimes (1, b, c) = (1, 0, 0)$$

► $\mathbb{T}_1^{(2)}$ is a group with respect to the operation $\otimes$.

► A rough path $(X, \mathbb{X})$ can be thought of as representing the increments of the group-valued path $t \mapsto (1, X_{0,t}, \mathbb{X}_{0,t}) =: \mathcal{X}_{0,t}$:

$$\mathcal{X}_{s,t} = \mathcal{X}_{0,s}^{-1} \otimes \mathcal{X}_{0,t}.$$

- Using the identifying

$$1 \leftrightarrow (1, 0, 0), \quad b \leftrightarrow (0, b, 0), \quad c \leftrightarrow (0, 0, c),$$

and the "power series"

$$\log(1 + x) = x - \frac{x^2}{2} + \cdots, \quad \exp(x) = 1 + x + \frac{x^2}{2!} + \cdots,$$

we define the logarithmic map and exponential map as

$$\log(1 + b + c) := b + c - \frac{1}{2}(b + c) \otimes (b + c) = b + c - \frac{1}{2}b \otimes b,$$

$$\exp(b + c) := 1 + (b + c) + \frac{1}{2}(b + c) \otimes (b + c) = 1 + b + c + \frac{1}{2}b \otimes b.$$

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$$\begin{aligned}T_0^{(2)}(\mathbb{R}^d) &= \{0\} \oplus \mathbb{R}^d \oplus (\mathbb{R}^d \otimes \mathbb{R}^d) \cong \mathbb{R}^d \oplus \mathbb{R}^{d \times d}, \\ T_1^{(2)}(\mathbb{R}^d) &\cong \exp(\mathbb{R}^d \oplus \mathbb{R}^{d \times d}).\end{aligned}$$

Rough paths as Lie-group valued paths

► $T_0^{(2)}(\mathbb{R}^d) \cong \mathbb{R}^d \oplus \mathbb{R}^{d \times d}$, $T_1^{(2)}(\mathbb{R}^d) \cong \exp(\mathbb{R}^d \oplus \mathbb{R}^{d \times d})$:

$T_0^{(2)}(\mathbb{R}^d)$ is a Lie algebra with respect to the bracket

$$[b + c, b' + c'] = b \otimes b' - b' \otimes b.$$

► Let $\mathfrak{g}^{(2)} \subset T_0^{(2)}(\mathbb{R}^d)$ be the sub-algebra generated by elements of the form $(0, b, 0)$:

- $\mathfrak{g}^{(2)} = \mathbb{R}^d \oplus \mathfrak{so}(d) = \text{span}\{e_i\} \oplus \text{span}\{e_{ij} := [e_i, e_j], 1 \leq i < j \leq d\}$
- $\mathfrak{g}^{(2)}$ is closed under $[\cdot, \cdot]$

$G^{(2)}(\mathbb{R}^d) := \exp(\mathfrak{g}^{(2)})$ is a Lie sub-group of $T_1^{(2)}(\mathbb{R}^d)$.

:the step-2 nilpotent Lie group with d generators.

Remark. A rough path $(X, \mathbb{X}) \in \mathcal{C}^\alpha$ is geometric
if and only if the map $t \mapsto \mathcal{X}_t$ takes value in $G^{(2)}(\mathbb{R}^d)$.

- $G^{(2)}(\mathbb{R}^d)$ admits a natural *homogeneous Carnot-Carathéodory norm* $\|\cdot\|_C$ with the property, for $\mathbf{x} = b + c$,

$$\|\mathbf{x}\|_C \asymp |b| + |c|^{\frac{1}{2}},$$

where $\asymp$ Lipschitz equivalence with constants that may depend on the dimension d .

- It generates *homogeneous Carnot-Carathéodory metric* d_C such that

$$d_C(\mathbf{X}_s, \mathbf{X}_t) = \|\mathbf{X}\|_C \asymp |X_{s,t}| + |\mathbb{X}_{s,t}|^{\frac{1}{2}}.$$

► We define the *truncated signature* of a smooth path $\gamma : [0, 1] \rightarrow \mathbb{R}^d$ by

$$G^{(2)}(\mathbb{R}^d) \ni S^{(2)}(\gamma) = \left(1, \int_0^t d\gamma(t), \int_0^1 \int_0^t d\gamma(s) \otimes d\gamma(t) \right),$$

then we have

$$\|\mathbf{x}\|_C := \inf \left\{ \int_0^t |\dot{\gamma}(t)| dt : \gamma \in \mathcal{C}^1([0, 1]; \mathbb{R}^d), S^{(2)}(\gamma) = \mathbf{x} \right\}.$$

Conclusion

1. For $\alpha \in (\frac{1}{3}, \frac{1}{2}]$, the space $\mathcal{C}^\alpha([0, T], V)$ of α -Hölder rough paths is the space of pairs $(X, \mathbb{X}) \in \mathcal{C}^\alpha([0, T]; V) \oplus \mathcal{C}_2^{2\alpha}([0, T]^2; V \otimes V)$,

$$\|X\|_\alpha + \sqrt{\|\mathbb{X}\|_{2\alpha}} = \sup_{x \neq s \in [0, T]} \frac{|X_{s,t}|}{|t-s|^\alpha} + \left(\sup_{x \neq s \in [0, T]} \frac{|\mathbb{X}_{s,t}|}{|t-s|^{2\alpha}} \right)^{\frac{1}{2}} < \infty,$$

satisfying Chen's relation:

$$\mathbb{X}_{s,t} - \mathbb{X}_{s,u} - \mathbb{X}_{u,t} = X_{s,u} \otimes X_{u,t}, \quad \forall s, u, t \in [0, T].$$

2. The space of geometric rough paths is

$$\mathcal{C}_g^\alpha([0, T], V) := \left\{ (X, \mathbb{X}) \in \mathcal{C}^\alpha : \text{Sym}(\mathbb{X}_{s,t}) = \frac{1}{2} X_{s,t} \otimes X_{s,t} \right\}.$$

3. $(X, \mathbb{X}) \in \mathcal{C}_g^\alpha \iff$ the map $t \mapsto \mathcal{X}_t$ takes value in $G^{(2)}(\mathbb{R}^d)$.

Exercise 2.9. (Interpolation)

Assume that $\mathbf{X}^n \in \mathcal{C}^\beta$, for $\beta \in (\frac{1}{3}, \frac{1}{2}]$, with uniform bounds

$$\sup_n \|\mathbf{X}^n\|_\beta < \infty \quad \text{and} \quad \sup_n \|\mathbb{X}^n\|_{2\beta} < \infty$$

and pointwise convergence $X_{0,t}^n \rightarrow X_{0,t}$ and $\mathbb{X}_{0,t}^n \rightarrow \mathbb{X}_{0,t}$.

Show that this implies

$$\mathbf{X} \in \mathcal{C}^\beta, \quad \text{and} \quad \varrho_\alpha(\mathbf{X}^n, \mathbf{X}) \rightarrow \mathbf{0},$$

for every $\alpha \in (\frac{1}{3}, \beta)$.

THANK YOU FOR YOUR ATTENTION!